

Code for Assignment 1 C4M

Problem 1

(e)

```
In [12]: import string
import math

ex1 = "MATHEMATICSFORCRYPTOGRAPHY"
key1 = "LEMON"

def encrypt(plain_txt, key):
    #first we need to construct the key. This isn't the most efficient implementation but will do for now
    rep_count = len(plain_txt) // len(key)
    remainder = len(plain_txt) % len(key)
    new_key = rep_count * key + key[:remainder]

    #now the encrypting part
    encrypted_text = ""

    for i in range(len(plain_txt)):
        new_index = (ord(plain_txt[i]) + ord(new_key[i]) - 2*ord('A')) % 26
        encrypted_text += chr(new_index + ord('A'))

    return encrypted_text

print(encrypt(ex1, key1))
```

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